# Evaluating Perfect Square Roots and Cube Roots
## Answer Key

Grade 8 / Pre-Algebra   The Number System

**Quick Answers**

| | | |
|---|---|---|
| 1. 12 | 5. 7.75 | 9. — |
| 2. 3 | 6. 7 | 10. 0.6 |
| 3. 15 | 7. 13 | 11. 6 |
| 4. $\frac{7}{8}$ | 8. $\frac{2}{5}$ | 12. 4.64 |

**Common wrong answers**

1. *If they got 72: halved the radicand instead of taking the square root*
2. *If they got 9: divided the radicand by 3 instead of taking the cube root*
5. *If they got 30: halved the radicand instead of taking the square root*
6. *If they got 18.52: took the square root instead of the cube root*
8. *If they got 0.016: took the cube root of the numerator only and left the denominator alone*
10. *If they got 0.06: moved the decimal point one place instead of taking the square root, giving 0.06*
11. *If they got 14.7: took the square root of the volume instead of the cube root*
12. *If they got 33.33: divided by 3 instead of taking the cube root*

### 1. Evaluate $\sqrt{144}$.

**Step 1.** A square root asks for the factor used *twice*, so search the square row of the table rather than dividing.
**Step 2.** $12 \times 12 = 144$, so the root is exact. $\boxed{12}$
*Common wrong answer:* 72 — the radicand was halved. Halving is not rooting: $72 \times 72$ is nowhere near 144.

### 2. Evaluate $\sqrt[3]{27}$.

**Step 1.** A cube root asks for the factor used *three* times.
**Step 2.** $27 = 3 \times 3 \times 3$, so the cube root is 3. $\boxed{3}$
*Why not* 9*:* 9 is $27 \div 3$, and $9 \times 9 \times 9 = 729$, not 27. Dividing by three and taking a cube root are different operations.

### 3. Evaluate $\sqrt{225}$.

**Step 1.** 225 is past most memorised tables, so factor it into pieces that are perfect squares: $225 = 9 \times 25$.
**Step 2.** The square root of a product is the product of the roots: $\sqrt{9} \times \sqrt{25} = 3 \times 5$. $\boxed{15}$
*Check:* $15 \times 15 = 225$.

**4. Evaluate $\sqrt{\frac{49}{64}}$.**

**Step 1.** The root of a quotient is the quotient of the roots, so the numerator and denominator can be handled separately: $\sqrt{\frac{49}{64}} = \frac{\sqrt{49}}{\sqrt{64}}$.

**Step 2.** Both are perfect squares: $\sqrt{49} = 7$ and $\sqrt{64} = 8$. $\boxed{\frac{7}{8}}$

*Check:* $\frac{7}{8} \times \frac{7}{8} = \frac{49}{64}$. The answer stays a fraction — there is nothing to round.

**5. Locate and approximate $\sqrt{60}$.**

**(a)** 60 is not in the square row, so trap it between the two nearest entries: $49 < 60 < 64$, which gives $7 < \sqrt{60} < 8$.

**(b)** $\sqrt{60} = 7.74596\ldots$, which rounds to $\boxed{7.75}$

**(c)** The rounded value terminates, so $7.75 = \frac{31}{4}$ is rational, while $\sqrt{60}$ is irrational — its decimal never ends and never repeats. 7.75 is a stand-in, not an equality. (Accept any sentence making that distinction; $7.75^2 = 60.0625 \neq 60$ is also a complete answer.)

*Common wrong answer:* 30 — the radicand was halved.

**6. Evaluate $\sqrt[3]{343}$.**

**Step 1.** The index is 3, so the cube row is the right table — the square row would answer a different question.

**Step 2.** $343 = 7 \times 7 \times 7$, and checking, $7^2 = 49$ then $49 \times 7 = 343$. $\boxed{7}$

*Common wrong answer:* about 18.5 — $\sqrt{343}$ instead of $\sqrt[3]{343}$. The index tells you how many equal factors to look for.

**7. Square patio of area 169 square feet.**

**Step 1.** For a square, area $= s^2$, so $s^2 = 169$ and the side is the square root of the area — the exponent 2 in the area formula is what makes it a square root.

**Step 2.** $13 \times 13 = 169$, so $s = 13$. $\boxed{13 \text{ ft}}$

*Units:* the area was in square feet, and a length is in feet — taking the square root of an area returns a plain length unit.

**8. Evaluate $\sqrt[3]{\frac{8}{125}}$.**

**Step 1.** Handle the numerator and denominator separately, exactly as for a square root: $\sqrt[3]{\frac{8}{125}} = \frac{\sqrt[3]{8}}{\sqrt[3]{125}}$.

**Step 2.** $8 = 2 \times 2 \times 2$ and $125 = 5 \times 5 \times 5$, so the roots are 2 and 5. $\boxed{\frac{2}{5}}$

*Common wrong answer:* 0.016 — the cube root was taken on top only, leaving $\frac{2}{125}$. Whatever you do to the numerator you must do to the denominator.

**9. Compare $\sqrt{50}$ with 7.1.**

**Step 1.** Comparing a root with a decimal is easiest as a comparison of squares, because squaring removes the radical exactly: $(\sqrt{50})^2 = 50$ and $7.1^2 = 50.41$.

**Step 2.** $50 < 50.41$, and both numbers are positive, so the same order holds before squaring.

$$\boxed{\sqrt{50} \; < \; 7.1}$$

*Worth saying out loud:* $\sqrt{50} = 7.0710\ldots$, so a student who rounds to 7.1 first will conclude the two are equal. They are not — rounding early destroys the comparison.

### 10. Evaluate $\sqrt{0.36}$.

**Step 1.** Decimals hide their perfect squares, so rewrite as a fraction: $0.36 = \frac{36}{100}$.

**Step 2.** Both parts are perfect squares: $\frac{\sqrt{36}}{\sqrt{100}} = \frac{6}{10}$. $\boxed{0.6}$

*Check:* $0.6 \times 0.6 = 0.36$. *Common wrong answer:* 0.06 — the decimal point was shifted instead of a root taken. $0.06 \times 0.06 = 0.0036$, which is not 0.36.

### 11. Storage cube of volume 216 cubic centimetres.

**Step 1.** For a cube, volume $= e^3$, so $e^3 = 216$ and the edge is the *cube* root — the exponent in the formula sets the index of the root.

**Step 2.** From the cube row, $6 \times 6 \times 6 = 216$. $\boxed{6 \text{ cm}}$

*Common wrong answer:* about 14.7 — $\sqrt{216}$ was used. An area formula squares one length, a volume formula cubes it, so a volume undoes with a cube root.

### 12. Locate and approximate $\sqrt[3]{100}$.

**(a)** Use the cube row: $64 < 100 < 125$, so $4 < \sqrt[3]{100} < 5$.

**(b)** $\sqrt[3]{100} = 4.6415\ldots$, which rounds to $\boxed{4.64}$

**(c)** The claim is **false**. $\sqrt{100} = 10$ while $\sqrt[3]{100} \approx 4.64$, so the cube root is the *smaller* one. Splitting 100 into three equal factors gives smaller factors than splitting it into two: for any $n > 1$, the higher the index, the closer the root sits to 1. (Accept any answer that identifies the claim as false and compares 10 with 4.64.)

*Common wrong answer:* about 33.3 — $100 \div 3$ instead of a cube root.